

Flavoured CP-asymmetry cancellations with a real Casas–Ibarra matrix

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Abstract

We consider type-I seesaw leptogenesis with three heavy Majorana neutrinos and a real Casas–Ibarra matrix. We construct an analytic one-parameter family of real orthogonal matrices for which $\epsilon_{\tau\tau}^{(1)} = 0$. For vanishing Majorana phases, we give a numerical example with $\epsilon_{ee}^{(1)} = -\epsilon_{\mu\mu}^{(1)} \neq 0$. We also construct a solution for which all diagonal entries vanish while the off-diagonal coherent source remains non-zero. The notation follows Ref. [1].

1 Setup and notation

We work in the basis in which the charged-lepton Yukawa matrix and the heavy Majorana mass matrix are diagonal. Following Ref. [1], the radiatively corrected Casas–Ibarra (CI) parametrisation is

$$Y_{\alpha j} = \pm i \frac{\sqrt{2}}{v} U_{\alpha a} \sqrt{m_a} O_{ja} \sqrt{f^{-1}(M_j)}, \quad (1)$$

where repeated light-neutrino indices are summed, $O^T O = O O^T = \mathbf{1}$, and

$$f(M_j) \equiv M_j^{-1} \left[1 - \frac{M_j^2}{16\pi^2 v^2} \left(\frac{\log(M_j^2/m_H^2)}{M_j^2/m_H^2 - 1} + 3 \frac{\log(M_j^2/m_Z^2)}{M_j^2/m_Z^2 - 1} \right) \right]. \quad (2)$$

The Higgs convention is $\langle \Phi^{(0)} \rangle = v/\sqrt{2} \simeq 174$ GeV.

We take the CI angles to be real,

$$O = R_{23}(x_1) R_{13}(x_2) R_{12}(x_3), \quad y_1 = y_2 = y_3 = 0, \quad (3)$$

with the rotation matrices and signs as in Eq. (11) of Ref. [1].

2 The CP-asymmetry tensor for real O

The CP-asymmetry tensor is given by

$$\begin{aligned} \epsilon_{\alpha\beta}^{(j)} = \frac{3}{32\pi(Y^\dagger Y)_{jj}} \sum_{k \neq j} \left\{ i \left[Y_{\alpha j} Y_{\beta k}^* (Y^\dagger Y)_{kj} - Y_{\beta j}^* Y_{\alpha k} (Y^\dagger Y)_{jk} \right] \sqrt{X_k} f_1(X_k) \right. \\ \left. + i \left[Y_{\alpha j} Y_{\beta k}^* (Y^\dagger Y)_{jk} - Y_{\beta j}^* Y_{\alpha k} (Y^\dagger Y)_{kj} \right] f_2(X_k) \right\}, \end{aligned} \quad (4)$$

where the loop functions are

$$f_1(X) \equiv \frac{2}{3} \left[(1+X) \log \left(1 + \frac{1}{X} \right) - \frac{2-X}{1-X} \right], \quad f_2(X) \equiv \frac{2}{3(X-1)}. \quad (5)$$

We focus on the case of $j = 1$. For clarity, we denote the heavy-neutrino mass ratios by

$$X_k \equiv \frac{M_k^2}{M_1^2}, \quad (6)$$

rather than x_k , since $x_{1,2,3}$ already denote the real CI angles.

For real O , the matrix $Y^\dagger Y$ is real and symmetric. Consequently, Eq. (4) gives

$$\epsilon_{\alpha\alpha}^{(1)} = \frac{3}{16\pi(Y^\dagger Y)_{11}} \sum_{k=2,3} (Y^\dagger Y)_{1k} \operatorname{Im}(Y_{\alpha 1}^* Y_{\alpha k}) g_k. \quad (7)$$

where

$$g_k \equiv \sqrt{X_k} f_1(X_k) + f_2(X_k), \quad k = 2, 3. \quad (8)$$

In particular,

$$\sum_{\alpha} \operatorname{Im}(Y_{\alpha 1}^* Y_{\alpha k}) = \operatorname{Im}(Y^\dagger Y)_{1k} = 0, \quad (9)$$

and hence

$$\operatorname{Tr} \epsilon^{(1)} = \epsilon_{ee}^{(1)} + \epsilon_{\mu\mu}^{(1)} + \epsilon_{\tau\tau}^{(1)} = 0. \quad (10)$$

3 An exact analytic family with $\epsilon_{\tau\tau}^{(1)} = 0$

From Eq. (1), we have

$$(Y^\dagger Y)_{1k} = \frac{2}{v^2} \sqrt{f^{-1}(M_1) f^{-1}(M_k)} \sum_a m_a O_{1a} O_{ka}, \quad (11)$$

$$\operatorname{Im}(Y_{\alpha 1}^* Y_{\alpha k}) = \frac{2}{v^2} \sqrt{f^{-1}(M_1) f^{-1}(M_k)} \sum_{a \neq b} \sqrt{m_a m_b} O_{1a} O_{kb} \operatorname{Im}(U_{\alpha a}^* U_{\alpha b}). \quad (12)$$

Defining

$$w_k \equiv f^{-1}(M_k) g_k, \quad k = 2, 3, \quad (13)$$

$$\mathcal{A}_{1k} \equiv \sum_a m_a O_{1a} O_{ka}, \quad (14)$$

$$\mathcal{B}_{1k}^\alpha \equiv \sum_{a \neq b} \sqrt{m_a m_b} O_{1a} O_{kb} \operatorname{Im}(U_{\alpha a}^* U_{\alpha b}) \quad (15)$$

$$= \sum_{a < b} \sqrt{m_a m_b} (O_{1a} O_{kb} - O_{1b} O_{ka}) \operatorname{Im}(U_{\alpha a}^* U_{\alpha b}), \quad (16)$$

Eq. (7) can be rewritten as

$$\epsilon_{\alpha\alpha}^{(1)} = \frac{3}{8\pi v^2 \mathcal{A}_{11}} \sum_{k=2,3} w_k \mathcal{A}_{1k} \mathcal{B}_{1k}^\alpha \quad (17)$$

An explicit solution for $\epsilon_{\tau\tau}^{(1)} = 0$ can be obtained by setting $x_1 = 0$ while leaving x_3 free. With $c_i \equiv \cos x_i$ and $s_i \equiv \sin x_i$,

$$O = \begin{pmatrix} c_2 c_3 & c_2 s_3 & s_2 \\ -s_3 & c_3 & 0 \\ -s_2 c_3 & -s_2 s_3 & c_2 \end{pmatrix}. \quad (18)$$

This leads to

$$\mathcal{A}_{12} = c_2 c_3 s_3 \Delta_{21}, \quad (19)$$

$$\mathcal{A}_{13} = c_2 s_2 \Delta_3, \quad (20)$$

where

$$\Delta_{21} \equiv m_2 - m_1, \quad (21)$$

$$\Delta_3 \equiv m_3 - m_1 c_3^2 - m_2 s_3^2. \quad (22)$$

and

$$\mathcal{B}_{12}^\alpha = c_2 Q_{12}^\alpha + s_2 \mathcal{Q}_D^\alpha, \quad (23)$$

$$\mathcal{B}_{13}^\alpha = \mathcal{Q}_E^\alpha. \quad (24)$$

where

$$Q_{ab}^\alpha \equiv \sqrt{m_a m_b} \operatorname{Im}(U_{\alpha a}^* U_{\alpha b}), \quad a < b, \quad (25)$$

$$\mathcal{Q}_D^\alpha \equiv s_3 Q_{13}^\alpha - c_3 Q_{23}^\alpha, \quad (26)$$

$$\mathcal{Q}_E^\alpha \equiv c_3 Q_{13}^\alpha + s_3 Q_{23}^\alpha, \quad (27)$$

Thus, apart from the common non-zero prefactor in Eq. (17),

$$\epsilon_{\alpha\alpha}^{(1)} \propto c_2 \{w_2 c_3 s_3 \Delta_{21} (c_2 Q_{12}^\alpha + s_2 \mathcal{Q}_D^\alpha) + w_3 s_2 \Delta_3 \mathcal{Q}_E^\alpha\}. \quad (28)$$

For $c_2 \neq 0$, the condition $\epsilon_{\tau\tau}^{(1)} = 0$ can therefore be solved exactly as

$$\boxed{\tan x_2 = -\frac{w_2 c_3 s_3 \Delta_{21} Q_{12}^\tau}{w_2 c_3 s_3 \Delta_{21} \mathcal{Q}_D^\tau + w_3 \Delta_3 \mathcal{Q}_E^\tau}} \quad (29)$$

for a given x_3 . Equation (29) is the desired analytic cancellation condition for $\epsilon_{\tau\tau}^{(1)} = 0$. It defines a continuous one-parameter family in the (x_2, x_3) plane. Allowing x_1 to vary gives the expected larger solution surface in the full three-angle space.

4 Numerical example

As an illustration, consider NO and use the oscillation parameters adopted in Ref. [1],

$$\theta_{12} = 33.68^\circ, \quad \theta_{13} = 8.56^\circ, \quad \theta_{23} = 43.30^\circ, \quad \delta = 212^\circ, \quad \alpha_{21} = \alpha_{31} = 0, \quad (30)$$

together with

$$m_1 = 0.010 \text{ eV}, \quad \Delta m_{21}^2 = 7.49 \times 10^{-5} \text{ eV}^2, \quad \Delta m_{31}^2 = 2.513 \times 10^{-3} \text{ eV}^2. \quad (31)$$

For illustration, choose

$$M_1 = 10^8 \text{ GeV}, \quad M_2 = 3M_1, \quad M_3 = 10M_1, \quad (32)$$

and evaluate both $f(M_j)$ and the loop functions in Eq. (5). Taking

$$x_1 = 0, \quad x_3 = \frac{\pi}{4}, \quad (33)$$

Eq. (29) gives

$$x_2 = 0.0230176 = 1.31881^\circ. \quad (34)$$

The corresponding real orthogonal matrix is

$$O = \begin{pmatrix} 0.706919474 & 0.706919474 & 0.023015524 \\ -0.707106781 & 0.707106781 & 0 \\ -0.016274433 & -0.016274433 & 0.999735108 \end{pmatrix}. \quad (35)$$

Substitution into the CP-asymmetry tensor gives

$$\left(\epsilon_{ee}^{(1)}, \epsilon_{\mu\mu}^{(1)}, \epsilon_{\tau\tau}^{(1)} \right) = (3.5313 \times 10^{-11}, -3.5313 \times 10^{-11}, \mathcal{O}(10^{-26})). \quad (36)$$

Both $\epsilon_{\tau\tau}^{(1)} = 0$ and $\text{Tr } \epsilon^{(1)} = 0$ are thus satisfied up to numerical precision, while the electron and muon asymmetries are individually non-zero.

5 A purely off-diagonal coherent source

We next ask whether a stronger condition can be realised,

$$\epsilon_{\mu\mu}^{(1)} = \epsilon_{\tau\tau}^{(1)} = 0, \quad \epsilon_{\alpha\beta}^{(1)} \neq 0 \quad \text{for some } \alpha \neq \beta. \quad (37)$$

Equation (10) immediately implies that $\epsilon_{ee}^{(1)} = 0$ as well. Thus the question is whether the source tensor can be non-zero while having a completely vanishing diagonal in the charged-lepton flavour basis. The answer is yes, provided the loop weights satisfy an appropriate non-degeneracy condition.

To make the structure manifest, define three real coefficients

$$\mathcal{C}_{ab} \equiv \sqrt{m_a m_b} \sum_{k=2,3} w_k \mathcal{A}_{1k} (O_{1a} O_{kb} - O_{1b} O_{ka}), \quad a < b. \quad (38)$$

The full CP-asymmetry tensor, including its off-diagonal entries, can then be written as

$$\epsilon_{\alpha\beta}^{(1)} = \frac{3i}{16\pi v^2 \mathcal{A}_{11}} \sum_{a < b} \mathcal{C}_{ab} (U_{\alpha a} U_{\beta b}^* - U_{\alpha b} U_{\beta a}^*). \quad (39)$$

For $\alpha = \beta$, this reduces to Eq. (17).

For the remainder of this section, we set $\alpha_{21} = \alpha_{31} = 0$. Substituting the PMNS matrix into Eq. (39), the electron-flavour diagonal entry becomes

$$\epsilon_{ee}^{(1)} = -\frac{3c_{13}s_{13}\sin\delta}{8\pi v^2 \mathcal{A}_{11}} (c_{12}\mathcal{C}_{13} + s_{12}\mathcal{C}_{23}). \quad (40)$$

For the physical mixing angles and $\sin\delta \neq 0$, it follows that

$$\epsilon_{ee}^{(1)} = 0 \quad \Longleftrightarrow \quad c_{12}\mathcal{C}_{13} + s_{12}\mathcal{C}_{23} = 0. \quad (41)$$

The τ -flavour diagonal entry is

$$\epsilon_{\tau\tau}^{(1)} = \frac{3s_{13}c_{23}\sin\delta}{8\pi v^2 \mathcal{A}_{11}} \left[-s_{23}\mathcal{C}_{12} + c_{23}c_{13} (c_{12}\mathcal{C}_{13} + s_{12}\mathcal{C}_{23}) \right]. \quad (42)$$

Once Eq. (41) is imposed, this reduces to

$$\epsilon_{\tau\tau}^{(1)} = -\frac{3s_{13}s_{23}c_{23}\sin\delta}{8\pi v^2 \mathcal{A}_{11}} \mathcal{C}_{12}. \quad (43)$$

Thus, again for the physical mixing angles and $\sin\delta \neq 0$,

$$\epsilon_{\tau\tau}^{(1)} = 0 \quad \Longleftrightarrow \quad \mathcal{C}_{12} = 0. \quad (44)$$

Finally, Eq. (10) implies that $\epsilon_{\mu\mu}^{(1)} = 0$ whenever $\epsilon_{ee}^{(1)} = \epsilon_{\tau\tau}^{(1)} = 0$. Therefore,

$$\boxed{\epsilon_{ee}^{(1)} = \epsilon_{\mu\mu}^{(1)} = \epsilon_{\tau\tau}^{(1)} = 0 \iff \mathcal{C}_{12} = 0, \text{ and } c_{12}\mathcal{C}_{13} + s_{12}\mathcal{C}_{23} = 0.} \quad (45)$$

A non-trivial solution therefore has the form

$$(\mathcal{C}_{12}, \mathcal{C}_{13}, \mathcal{C}_{23}) = \lambda(0, s_{12}, -c_{12}), \quad \lambda \neq 0. \quad (46)$$

Substituting Eq. (46) into Eq. (39) first gives

$$\epsilon_{\alpha\beta}^{(1)} = \frac{3i\lambda}{16\pi v^2 \mathcal{A}_{11}} \left\{ s_{12} (U_{\alpha 1} U_{\beta 3}^* - U_{\alpha 3} U_{\beta 1}^*) - c_{12} (U_{\alpha 2} U_{\beta 3}^* - U_{\alpha 3} U_{\beta 2}^*) \right\}. \quad (47)$$

For vanishing Majorana phases, the PMNS elements satisfy

$$\begin{pmatrix} s_{12}U_{e1} - c_{12}U_{e2} \\ s_{12}U_{\mu 1} - c_{12}U_{\mu 2} \\ s_{12}U_{\tau 1} - c_{12}U_{\tau 2} \end{pmatrix} = \begin{pmatrix} 0 \\ -c_{23} \\ s_{23} \end{pmatrix}, \quad (48)$$

while the third column is

$$\begin{pmatrix} U_{e3} \\ U_{\mu 3} \\ U_{\tau 3} \end{pmatrix} = \begin{pmatrix} s_{13}e^{-i\delta} \\ s_{23}c_{13} \\ c_{23}c_{13} \end{pmatrix}. \quad (49)$$

Using these expressions in Eq. (47) yields

$$\epsilon^{(1)} = \frac{3i\lambda}{16\pi v^2 \mathcal{A}_{11}} \begin{pmatrix} 0 & c_{23}s_{13}e^{-i\delta} & -s_{23}s_{13}e^{-i\delta} \\ -c_{23}s_{13}e^{i\delta} & 0 & -c_{13} \\ s_{23}s_{13}e^{i\delta} & c_{13} & 0 \end{pmatrix}. \quad (50)$$

Thus the diagonal vanishes identically, while in particular

$$\epsilon_{\mu\tau}^{(1)} = -\frac{3i\lambda c_{13}}{16\pi v^2 \mathcal{A}_{11}} \neq 0. \quad (51)$$

Notice that the tensor is Hermitian, as required.

We now give a constructive real- O solution. The definition (38) directly implies the identity

$$\boxed{\sqrt{m_1}\mathcal{C}_{23}O_{11} - \sqrt{m_2}\mathcal{C}_{13}O_{12} + \sqrt{m_3}\mathcal{C}_{12}O_{13} = 0.} \quad (52)$$

Indeed, direct substitution gives

$$\begin{aligned} & \sqrt{m_1}\mathcal{C}_{23}O_{11} - \sqrt{m_2}\mathcal{C}_{13}O_{12} + \sqrt{m_3}\mathcal{C}_{12}O_{13} \\ &= \sqrt{m_1 m_2 m_3} \sum_{k=2,3} w_k \mathcal{A}_{1k} \left[O_{11}(O_{12}O_{k3} - O_{13}O_{k2}) \right. \\ & \quad \left. - O_{12}(O_{11}O_{k3} - O_{13}O_{k1}) + O_{13}(O_{11}O_{k2} - O_{12}O_{k1}) \right] = 0, \end{aligned} \quad (53)$$

where the six terms in the square brackets cancel pairwise.

Substituting the non-trivial direction Eq. (46) into Eq. (52) gives

$$-\lambda(\sqrt{m_1}c_{12}O_{11} + \sqrt{m_2}s_{12}O_{12}) = 0. \quad (54)$$

Since $\lambda \neq 0$, this is equivalent to

$$\sqrt{m_1}c_{12}O_{11} + \sqrt{m_2}s_{12}O_{12} = 0. \quad (55)$$

Since the first row of the CI matrix in Eq. (3) is $(c_2 c_3, c_2 s_3, s_2)$, for $c_2 \neq 0$ this condition fixes

$$\boxed{\tan x_3 = -\sqrt{\frac{m_1}{m_2}} \cot \theta_{12}} \quad (56)$$

With x_3 fixed by Eq. (56), it remains to impose $\mathcal{C}_{12} = 0$. We now solve this condition for x_1 . Define

$$A \equiv c_2 c_3 s_3 \Delta_{21}, \quad B \equiv c_2 s_2 \Delta_3(x_3), \quad t_1 \equiv \tan x_1. \quad (57)$$

A direct evaluation of Eq. (38) gives

$$\mathcal{C}_{12} = \frac{\sqrt{m_1 m_2} c_2}{1 + t_1^2} [A(w_2 + w_3 t_1^2) + B(w_2 - w_3) t_1]. \quad (58)$$

Hence a non-trivial solution is obtained from

$$\boxed{t_1 = \frac{-B(w_2 - w_3) \pm \sqrt{B^2(w_2 - w_3)^2 - 4A^2 w_2 w_3}}{2Aw_3}} \quad (59)$$

provided

$$B^2(w_2 - w_3)^2 \geq 4A^2 w_2 w_3. \quad (60)$$

Equations (56) and (59), with x_2 chosen so that Eq. (60) holds, give an analytic construction of a purely off-diagonal CP source.

This possibility disappears when the two loop weights become equal. Indeed, if $w_2 = w_3 \equiv w$, orthogonality of O gives

$$\mathcal{C}_{ab} = w \sqrt{m_a m_b} (m_b - m_a) O_{1a} O_{1b}. \quad (61)$$

For non-zero, non-degenerate light-neutrino masses, $\mathcal{C}_{12} = 0$ then requires either $O_{11} = 0$ or $O_{12} = 0$. The first choice also sets $\mathcal{C}_{13} = 0$, while the second sets $\mathcal{C}_{23} = 0$; either case is incompatible with Eq. (46) unless $\lambda = 0$. Therefore a non-trivial coherent source requires $w_2 \neq w_3$ and the discriminant condition (60).

For a numerical check, use the same neutrino and heavy-neutrino parameters as in the preceding section. Taking

$$x_2 = \frac{\pi}{4}, \quad x_3 = -0.916897865, \quad x_1 = 0.333732583, \quad (62)$$

where x_3 and x_1 follow from Eqs. (56) and (59), gives

$$O = \begin{pmatrix} 0.430122661 & -0.561243705 & 0.707106781 \\ 0.609029909 & 0.758571605 & 0.231628343 \\ -0.666391076 & 0.331020579 & 0.668093040 \end{pmatrix}. \quad (63)$$

The resulting tensor is

$$\epsilon^{(1)} \simeq 10^{-9} \begin{pmatrix} 0 & -0.110030 - 0.176085i & 0.103687 + 0.165934i \\ -0.110030 + 0.176085i & 0 & -1.895430i \\ 0.103687 - 0.165934i & 1.895430i & 0 \end{pmatrix}, \quad (64)$$

where the three diagonal entries are below 6×10^{-26} in absolute value before rounding. This explicitly verifies

$$\epsilon_{ee}^{(1)} = \epsilon_{\mu\mu}^{(1)} = \epsilon_{\tau\tau}^{(1)} = 0, \quad \epsilon_{\alpha\beta}^{(1)} \neq 0 \quad (\alpha \neq \beta). \quad (65)$$

References

- [1] A. Granelli, K. Hamaguchi, M. E. Ramirez-Quezada, K. Shimada, J. Wada and T. Yokoyama, “Insights on the Scale of Leptogenesis from Neutrino Masses and Neutrinoless Double-Beta Decay,” *Eur. Phys. J. C* **85** (2025) 778, arXiv:2502.10093 [hep-ph].